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Gravity type water-purifying device

Abstract

A gravity type water-purifying device is provided. A gravity type water-purifying device according to an embodiment of the present invention comprises: a filtered water generating part comprising a filler module for filtering out foreign materials contained in raw water by using gravity, a housing which stores a predetermined amount of the raw water and has a filtering space formed therein in which the filter module is disposed, and an outlet formed through the housing to allow the filtered water generated through the filter module to be discharged to the outside; and a filtered water storing part which has a filtered water storing space formed therein in order to store a predetermined amount of filtered water generated by the filtered water generating part, is detachably connected to the filtered water generating part by unit of a connection pipe so as to be separable from the filtered water generating part and be movable by itself, and is provided with an opening/closing unit for discharging filtered water stored in the filtered water storing space to the outside.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
419023	12/1889	Roberts	210/313	B01D 24/183
445223	12/1890	Knight	55/501	B01D 37/025
513802	12/1893	Luscombe	210/476	C02F 1/001
541970	12/1894	Harder	210/314	C02F 1/001
546335	12/1894	Knight	210/458	B01D 27/02
548487	12/1894	Way	210/411	A61M 1/3633
569382	12/1895	Knight	210/123	B01D 39/1623
589887	12/1896	Ziegler	210/486	F25D 3/04
630846	12/1898	Bliss	210/462	B01D 29/33
645500	12/1899	Ziegler	210/473	B01D 25/26
676755	12/1900	Melcher	210/473	C02F 1/003
963324	12/1909	Randall	210/489	B01D 21/26
1276665	12/1917	Leopold	210/292	C02F 1/001
1401199	12/1920	Simpson	210/505	B01J 20/00
1531095	12/1924	Hoehn	210/346	B01D 29/39
1674203	12/1927	Holz	210/489	C02F 1/001
1723564	12/1928	Lewis	210/287	C02F 1/003
1726035	12/1928	Bomonti	210/486	B01D 29/41

1919448	12/1932	Norquist	210/313	B01D 29/66
1982924	12/1933	Norquist	210/486	B01D 29/39
2002826	12/1934	Moran	210/346	B01D 29/39
2028466	12/1935	Moran	210/486	B01D 29/39
2041495	12/1935	Schwiers	210/420	B01D 29/66
2190305	12/1939	Auberschek	210/441	B01D 35/04
2306986	12/1941	Tolman	210/486	B01D 39/2003
2426618	12/1946	Klein	210/486	B01D 25/26
2443087	12/1947	Ulrich	210/486	B01D 29/39
2468296	12/1948	Jacobowitz	210/486	B01D 29/39
2788901	12/1956	Boeddinghaus	210/486	B01D 29/111
2909285	12/1958	Besler	210/330	B01D 29/94
3295687	12/1966	Schmerler	55/379	B01D 29/39
3339742	12/1966	Kracklauer	210/232	B01D 29/39
3339743	12/1966	Bealle	210/256	B01J 47/012
3392837	12/1967	Sanzenbacher	210/282	C02F 1/003
3430769	12/1968	Sanzenbacher	210/282	C02F 1/003
3623614	12/1970	Schmidt, Jr.	210/486	B01D 29/111
4968423	12/1989	McKale	210/486	B01D 29/43
5035799	12/1990	Rosberg	210/411	B01D 29/66
5128036	12/1991	Svensson	210/264	C02F 1/003
5476526	12/1994	Attermeyer	55/496	B01D 46/0002
5562824	12/1995	Magnusson	210/290	C02F 1/003
5928506	12/1998	Bae	210/123	C02F 1/003
6013180	12/1999	Wang	210/255	C02F 1/003
6129841	12/1999	Dann	210/260	C02F 1/003
6248244	12/2000	Dann	210/764	C02F 1/003
D446791	12/2000	Beckham	D15/1	N/A
6331246	12/2000	Beckham	210/136	B01D 35/30
6419821	12/2001	Gadgil	210/256	B01D 39/2068
6602425	12/2002	Gadgil	210/744	B01D 39/2068
7014759	12/2005	Radford	141/330	C02F 1/003
7135114	12/2005	Nonninger	55/482	B01D 63/0821
7163618	12/2006	Beckham	210/473	B01D 35/153
7232517	12/2006	Shepherd	250/435	C02F 1/003
7306723	12/2006	Radford	141/330	C02F 1/003
7906019	12/2010	Elliott	210/205	A01N 59/16
8043502	12/2010	Nauta	210/473	C02F 1/003
8216465	12/2011	Nauta	210/473	C02F 1/003
9193605	12/2014	Minton-Edison	N/A	C02F 1/002
9352252	12/2015	Vestergaard Frandsen	N/A	B01D 29/62

9352979	12/2015	Kuennen	N/A	C02F 3/06
9975068	12/2017	Maresh	N/A	B01D 24/00
10336639	12/2018	Kuennen	N/A	C02F 1/281
10589195	12/2019	Han	N/A	C02F 1/004
10589199	12/2019	See	N/A	C02F 1/444
10589200	12/2019	Maresh	N/A	C02F 1/001
10647592	12/2019	Geelhood	N/A	C02F 1/002
10905976	12/2020	Johnson	N/A	B01D 21/0012
11161062	12/2020	Johnson	N/A	C02F 9/20
11202975	12/2020	Han	N/A	C02F 1/004
11247153	12/2021	Han	N/A	C02F 1/44
11872506	12/2023	Weitz	N/A	B01D 24/38
11938424	12/2023	Han	N/A	C02F 1/003
12311292	12/2024	Song	N/A	B01D 29/15
12318720	12/2024	Johnson	N/A	B01D 29/58
2002/0113022	12/2001	Gadgil	210/748.11	B01D 39/2068
2003/0213733	12/2002	Beckham	210/500.1	A61M 1/79
2004/0134932	12/2003	Lobdell	222/189.06	B67D 3/0029
2006/0144781	12/2005	Carlson	210/473	B01D 61/20
2007/0080118	12/2006	Lindsley	210/791	B01D 35/0276
2008/0217252	12/2007	Elliott	210/661	A01N 59/16
2011/0303589	12/2010	Kuennen	210/120	C02F 1/52
2012/0187052	12/2011	Elliott	210/764	A01N 59/16
2012/0267314	12/2011	Minton-Edison	210/695	C02F 1/44
2014/0144829	12/2013	Takeda	210/295	B01D 63/024
2014/0216993	12/2013	Pradeep	210/123	B01D 29/00
2015/0284260	12/2014	Huda	210/128	C02F 1/003
2016/0114271	12/2015	Michaud	210/807	C02F 1/003
2016/0236958	12/2015	Kuennen	N/A	C02F 1/283
2018/0154290	12/2017	See	N/A	C02F 1/002
2019/0209972	12/2018	Han	N/A	B01D 63/0821
2019/0247772	12/2018	Han	N/A	B01D 69/06
2020/0071189	12/2019	Geelhood	N/A	C02F 1/283
2020/0215463	12/2019	Johnson	N/A	B01D 35/027
2022/0047975	12/2021	Johnson	N/A	B01D 35/02
2022/0054958	12/2021	Han	N/A	B01D 36/001
2023/0166203	12/2022	Song	210/455	C02F 1/001
2024/0066450	12/2023	Lee	N/A	B01D 29/00

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
------------	------------------	---------	-----

200295215	12/2001	KR	N/A
20040042180	12/2003	KR	N/A
20090129389	12/2008	KR	N/A
1020150011745	12/2014	KR	N/A
101516969	12/2014	KR	N/A
20180116667	12/2017	KR	N/A
20200006135	12/2019	KR	N/A
20200071418	12/2019	KR	N/A

OTHER PUBLICATIONS

International Search Report issued in PCT/KR2021/019272 dated Mar. 24, 2022, 5 pgs. cited by applicant

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION

(1) This application is a 35 U.S.C. 371 National Phase Entry Application from PCT/KR2021/019272 filed Dec. 17, 2021, which claims priority to and the benefit of Korean Patent Application No. 10-2020-0183243, filed on Dec. 24, 2020, the disclosures of which are incorporated herein by reference in their entirety.

TECHNICAL FIELD

(2) The present invention relates to a gravity-type water-purifying device.

BACKGROUND ART

(3) Supplying clean and safe water is essentially necessary for health and survival of humans. However, since clean water is not sufficiently supplied in Africa or developing countries, people frequently use water in a contaminated pond or puddle as drinking water.

(4) When drinking contaminated water, the drinker may get various infectious diseases due to microbes included in the contaminated water. That is, there is a problem that the microbes contained in the contaminated water cause illness such as abdominal pain, diarrhea, vomiting and the like, and in a severe case threaten the life of the drinker.

(5) Accordingly, in order to prevent such problem, it is important to ensure that the food or water we eat is not contaminated by microbes.

(6) Currently, since the overall mortality rate can be reduced merely if clean water and sufficient food are secured in developing countries, it can be said that the effect of environmental improvement, which can be achieved by supplying clean water, is far greater than advances in medicine or treatment technology.

(7) However, since a conventional water purifier passes raw water through a filter by applying a certain water pressure to filter out foreign substances contained in the raw water, the purifier requires a pump facility capable of imparting the certain water pressure, and electrical facilities for operating the pump facility must also be provided. In the absence of such electrical facilities, there is a limitation that the conventional water purifier cannot be used.

DISCLOSURE

Technical Problem

(8) The present invention has been devised in view of the above points, and an object of the present

invention is to provide a gravity-type water-purifying device capable of conveniently producing filtered water by using gravity.

(9) Another object of the present invention is to provide a gravity-type water-purifying device capable of reducing downtime by configuring a storage tank in which filtered water is stored in a movable manner so that it can be individually separated.

Technical Solution

(10) In order to achieve the above objects, the present invention provides a gravity-type water-purifying device comprising: a filtered water production part including: a filter module for filtering out foreign substances contained in raw water by using gravity, a housing in which a filtration space is formed to store a certain amount of the raw water and to dispose the filter module therein, and an outflux formed in the housing to allow the filtered water produced through the filter module to be discharged to the outside; and a movable filtered water storage part in which a filtered water storage space is formed to store a certain amount of the filtered water produced from the filtered water production part, wherein the filtered water storage part is detachably connected to the filtered water production part through a connection pipe so as to be separable from the filtered water production part and be movable by itself, and is provided with an opening/closing unit for discharging the filtered water stored in the filtered water storage space to the outside.

(11) In addition, the movable filtered water storage part may include a box-shaped storage tank including the filtered water storage space, and a fitting formed in the storage tank so as to be detachably coupled to an end of the connection pipe.

(12) In addition, the connection pipe may further include a filter for re-filtering the filtered water moving from the outflux to the movable filtered water storage part.

(13) In addition, the connection pipe may further include an opening/closing valve for allowing or blocking the movement of the filtered water from the outflux to the movable filtered water storage part.

(14) In addition, the filter module may comprise: a plurality of plate-shaped filter members, which are formed in a plate shape having a predetermined area, produce filtered water from raw water introduced from the outside by gravity, and are spaced apart from each other in parallel along one direction while one surface of each filter member faces each other at a predetermined interval; and a filtered water integration member for fixing one side of each of the plurality of plate-shaped filter members so that the plurality of plate-shaped filter members can be kept spaced apart from each other along one direction while facing each other, and for integrating and discharging the filtered water individually produced from each of the plurality of plate-shaped filter members.

(15) In addition, each of the plurality of plate-shaped filter members may include: a support frame comprising a loop-shaped edge member having an empty space open on both sides, and a partition member coupled to the edge member to partition the empty space into a plurality of storage spaces; a pair of filtration members formed in a plate shape having a predetermined area and coupled to both surfaces of the support frame to produce filtered water from raw water moving from the outside to the plurality of storage spaces; at least one communication path formed in the partition member to allow the plurality of storage spaces to communicate with each other; and a receiving port provided on one side of the support frame so as to communicate with any one of the plurality of storage spaces and connected to the filtered water integration member via a tube.

(16) In addition, the communication path may be an accommodation groove formed to be recessed into the partition member.

(17) In addition, the edge member may include a stepped portion formed along the inner side of the edge, and an edge side of the filtration member may be fixed to the stepped portion.

(18) In addition, the partition member may include at least one lateral member having a predetermined length and at least one longitudinal member disposed to intersect the lateral member.

(19) In addition, the filtration member may include a plate-shaped support having a predetermined area, and a nanofiber web layer formed of nanofiber and disposed on one surface or both surfaces

of the support.

(20) In addition, all of the plurality of plate-shaped filter members may be provided with the same filter member.

(21) In addition, the filtered water integration member may include: a body having an integration space for integrating the filtered water incoming from each of the plurality of plate-shaped filter members; a plurality of protrusions protruding from one surface of the body to form a plurality of accommodation grooves into which one side of the plurality of plate-shaped filter members are respectively inserted; and a fastening member fastened to the body and passing through all of the plurality of protrusions so as to prevent each of the plate-shaped filter members respectively inserted into the plurality of accommodation grooves from being separated from the accommodation grooves.

(22) In addition, the body may further include a plurality of inlets connected one-to-one to the receiving ports through a tube wherein the receiving ports are respectively provided in the plurality of plate-shaped filter members so as to introduce the filtered water individually produced by the plurality of plate-shaped filter members into the integration space, and at least one outlet for discharging the filtered water from the integration space to the outside.

Advantageous Effects

(23) According to the present invention, filtered water can be simply produced by using gravity even under poor conditions in which power such as electricity is not supplied, thereby obtaining clean filtered water even in a poor environment. As such, it is possible to simply obtain filtered water even in a poor environment, thereby preventing diseases caused by ingestion of contaminated water and improving the quality of life.

(24) In addition, according to the present invention, a storage tank in which filtered water is stored is configured in a movable manner so that it can be individually separated, whereby downtime can be reduced to increase the production of the filtered water.

Description

DESCRIPTION OF DRAWINGS

(1) FIG. 1 is a view showing a gravity-type water-purifying device according to an embodiment of the present invention.

(2) FIG. 2 is a view showing a state in which a movable filtered water storage part in FIG. 1 is separated.

(3) FIG. 3 is a front view showing a housing and a movable filtered water storage part in FIG. 1 partially cut away.

(4) FIG. 4 is a view showing a connection relationship between a connection pipe and a storage tank that can be applied to a gravity-type water-purifying device according to an embodiment of the present invention.

(5) FIG. 5 is a view showing a filter module extracted from FIG. 1.

(6) FIG. 6 is an exploded view of FIG. 5.

(7) FIG. 7 is a view showing a plate-shaped filter member applicable to a filter module in a gravity-type water-purifying device according to an embodiment of the present invention.

(8) FIG. 8 is a view showing a state in which the filtration member in FIG. 7 is separated.

(9) FIG. 9 is a cross-sectional view taken along line A-A of FIG. 7.

(10) FIG. 10 is a cross-sectional view taken along line B-B of FIG. 7.

(11) FIG. 11 is an enlarged view of part C in FIG. 8.

(12) FIG. 12 is a view showing a filtered water integration member applicable to a filter module in a gravity-type water-purifying device according to an embodiment of the present invention.

(13) FIG. 13 is a view showing a state in which a part in FIG. 12 is cut away.

(14) FIG. 14 is a view showing a coupling relationship between a filtered water integration member and a plate-shaped filter member in a gravity-type water-purifying device according to an embodiment of the present invention.

(15) FIG. 15 is a schematic diagram illustrating a detailed configuration of a filtration member applicable to a plate-shaped filter member in a gravity-type water-purifying device according to an embodiment of the present invention.

MODES OF THE INVENTION

(16) Hereinafter, with reference to the accompanying drawings, embodiments of the present invention will be described in detail so as to be easily implemented by one of ordinary skill in the art to which the present invention pertains. The present invention may be embodied in a variety of forms and is not be limited to the embodiments described herein. In order to clearly describe the present invention in the drawing, parts irrelevant to the description are omitted from the drawings; and throughout the specification, same or similar components will be referred to as like reference numerals.

(17) A gravity-type water-purifying device **1000** according to an embodiment of the present invention may produce filtered water from raw water by using water pressure due to gravity or potential energy.

(18) Herein, the filtered water may be used as potable water or drinking water.

(19) In this case, the gravity-type water-purifying device **1000** according to an embodiment of the present invention may include a filter module **100** for producing filtered water from raw water, and the filtered water produced from the filter module **100** may be moved toward a storage tank **1210** through a connection pipe **1300** and be stored in the storage tank **1210**. In addition, the storage tank **1210** may be detachably coupled to the connection pipe **1300**.

(20) Accordingly, in the gravity-type water-purifying device **1000** according to an embodiment of the present invention, the storage tank **1210** may be easily separated from the connection pipe **1300** to be replaced with another storage tank **1210**.

(21) That is, in the gravity-type water-purifying device **1000** according to an embodiment of the present invention, when the storage tank **1210** is full of the filtered water or the storage tank **1210** needs to be washed, the storage tank **1210** connected to the connection pipe **1300** may be easily replaced with another storage tank.

(22) Through this, in the gravity-type water-purifying device **1000** according to an embodiment of the present invention, downtime during which the filtered water cannot be produced may be reduced, and contamination of the filtered water that may occur in the process of transferring the filtered water produced from the filter module **100** to another storage container may be prevented in advance.

(23) To this end, the gravity-type water-purifying device **1000** according to an embodiment of the present invention includes a filtered water production part **1100** and a movable filtered water storage part **1200** as shown in FIGS. 1 to 3, and the filtered water production part **1100** may be connected to the movable filtered water storage part **1200** via the connection pipe **1300**.

(24) The filtered water production part **1100** may produce the filtered water from raw water by using gravity as described above.

(25) To this end, the filtered water production part **1100** may include a filter module **100** and a housing **200**.

(26) Specifically, the housing **200** may include a filtration space **210** in which raw water to be treated is stored, and the filter module **100** may be disposed in the filtration space **210**.

(27) For example, as shown in FIG. 3, the housing **200** may be formed in a box shape having an internal space of a predetermined volume, and the internal space may be the filtration space **210** in which a certain amount of the raw water is stored and the filter module **100** is disposed.

(28) Accordingly, when the filtration space **210** is filled with the raw water, at least a portion of the filter module **100** may be submerged in the raw water filled in the filtration space **210** as shown in

FIG. 3.

(29) In this case, the filtered water production part **1100** may include an outflux **220** for discharging the filtered water produced from the filter module **100** to the movable filtered water storage part **1200**.

(30) For example, the outflux **220** may be formed on the bottom surface of the housing **200**, and the outlet **127** of the filter module **100** may be coupled to the outflux **220**. In addition, the end of the connection pipe **1300** interconnecting the filtered water production part **1100** and the movable filtered water storage part **1200** may be coupled to the outflux **220**.

(31) In this case, the outflux **220** may serve both as a role of discharging the filtered water produced from the filter module **100** to the outside and as a fastener for fastening the outlet **127** of the filter module **100**.

(32) Accordingly, when the filtered water produced from the filter module **100** is discharged from the filtration space **210** through the outflux **220**, the filtered water may move to the movable filtered water storage part **1200** along the connection pipe **1300**. Accordingly, the water level of the raw water filled in the filtration space **210** may be lowered, and the pressure due to gravity generated as the water level of the raw water is lowered may be applied to the filter module **100**.

(33) As a result, the filter module **100** may produce filtered water by filtering out foreign substances from the raw water by using the gravity generated when the water level of the raw water changes. A detailed description of the filter module **100** will be provided later.

(34) As such, the gravity-type water-purifying device **1000** according to an embodiment of the present invention may produce filtered water by using the filter module **100** in the filtration space **210**, and the filtered water produced in the filtration space **210** by the filter module **100** may move to the movable filtered water storage part **1200** through the outflux **220**.

(35) As a result, the gravity-type water-purifying device **1000** according to an embodiment of the present invention may easily produce filtered water without using additional power such as electricity, whereby filtered water may be easily produced even in a harsh environment where infrastructure such as electricity is not installed or electricity is difficult to supply.

(36) In the drawings and description, it has been shown and described that the outflux **220** for discharging the filtered water produced from the filter module **100** to the movable filtered water storage part **1200** is formed on the bottom surface of the housing **200**, but the present invention is not limited thereto, and the position of the outflux **220** may be changed to an appropriate location according to design conditions in consideration of the connection relationship with the filter module **100**.

(37) For example, if the outflux **220** may move the filtered water produced from the filter module **100** to the movable filtered water storage part **1200** through the connection pipe **1300** without flowing out to the filtration space **210**, it may be formed in the housing **200** so as to be located in an appropriate position.

(38) In this case, the housing **200** may further include an inlet **230** for supplying the raw water to the filtration space **210**. In this case, as shown in FIG. 3, a filtering net **240** may be detachably coupled to the inlet **230**. Such a filtering net **240** may filter out bulky foreign substances such as fallen leaves or tree branches included in raw water when the raw water is supplied to the filtration space **210**.

(39) Meanwhile, the gravity-type water-purifying device **1000** according to an embodiment of the present invention may further include a support part **300** for supporting the housing **200**.

(40) Such a support part **300** may support one side of the housing **200** so that the housing **200** may be disposed in a state of being spaced apart from the bottom surface by a certain height.

(41) For example, the support part **300** may be a frame structure formed to have a predetermined height, and may support the lower surface of the housing **200**.

(42) Herein, the support part **300** may be configured separately from the housing **200**, or may be formed integrally with the housing **200**.

(43) As described above, the movable filtered water storage part **1200** may be connected to the filtered water production part **1100** via a connection pipe **1300** having one end connected to the outflux **220**.

(44) Accordingly, the movable filtered water storage part **1200** may store a certain amount of the filtered water produced from the filter module **100** therein.

(45) To this end, the movable filtered water storage part **1200** may include a storage tank **1210** having a filtered water storage space **1230** for storing a certain amount of the filtered water introduced from the filtered water production part **1100**, and the storage tank **1210** may include a fitting **1220** to which an end of the connection pipe **1300** is connected.

(46) Accordingly, the filtered water produced from the filtered water production part **1100** may be introduced into the filtered water storage space **1230** through the connection pipe **1300**.

(47) In this case, as shown in FIG. 4, an end of the connection pipe **1300** may be detachably coupled to the fitting **1220**.

(48) Accordingly, when the storage tank **1210** needs to be washed or a large amount of the filtered water is filled in the filtered water storage space **1230**, the user may easily replace the storage tank **1210** with another storage tank by separating the end of the connection pipe **1300** from the fitting **1220**.

(49) As a result, in the gravity-type water-purifying device **1000** according to an embodiment of the present invention, the storage tank **1210** may be easily replaced through the coupling and separation of the connection pipe **1300** and the fitting **1220**, whereby the filtered water may be continuously produced through the filter module **100**.

(50) That is, in the gravity-type water-purifying device **1000** according to an embodiment of the present invention, the storage tank **1210** may be individually separated and combined with the connection pipe **1300**, thereby reducing downtime during which the production of filtered water is stopped.

(51) In this case, as shown in FIG. 4, the storage tank **1210** may further include a stopper **1250** for opening and closing the fitting **1220**, and a handle **1260** for allowing the user to easily handle the storage tank **1210**.

(52) In addition, the storage tank **1210** may further include an opening/closing unit **1240** for discharging the filtered water filled in the filtered water storage space **1230** to the outside through a user's manipulation. For example, the opening/closing unit **1240** may be a well-known opening/closing valve.

(53) Accordingly, when a certain amount of filtered water is filled in the filtered water storage space **1230**, the user may separate the connection pipe **1300** from the fitting **1220** and then cover the fitting **1220** with the stopper **1250**.

(54) Through this, the stopper **1250** may prevent foreign substances from flowing into the filtered water storage space **1230** from the outside.

(55) In addition, the user may conveniently move the storage tank **1210** to a desired position by using the handle **1260** in a state in which the filtered water storage space **1230** is filled with a certain amount of filtered water.

(56) Further, when the user wants to use the filtered water, the filtered water stored in the filtered water storage space **1230** may be taken out by manipulating the opening/closing unit **1240**. Thus, the user may conveniently use the filtered water stored in the filtered water storage space **1230**.

(57) As described above, in the gravity-type water-purifying device **1000** according to an embodiment of the present invention, the movable filtered water storage part **1200** has the storage tank **1210** implemented as a portable water bottle type so that it may be moved and stored individually, thereby improving user convenience.

(58) In addition, when the user wants to use the filtered water produced from the filter module **100** in the gravity-type water-purifying device **1000** according to an embodiment of the present invention, the user may directly take out the filtered water stored in the filtered water storage space

1230 through the opening/closing unit **1240** provided in the storage tank **1210**.

(59) Accordingly, since it is not necessary to move the filtered water filled in the storage tank **1210** to a separate storage container for use of the filtered water, contamination of the filtered water that may occur in the process of moving the filtered water to a separate storage container may be prevented in advance.

(60) Meanwhile, the gravity-type water-purifying device **1000** according to an embodiment of the present invention may further include an opening/closing valve **1500** in the connection pipe **1300** for allowing or blocking the movement of filtered water produced from the filter module **100** toward the movable filtered water storage part **1200**.

(61) That is, when the storage tank **1210** needs to be replaced, the user may block the movement of filtered water produced from the filter module **100** toward the movable filtered water storage part **1200** along the connection pipe **1300** by manipulating the opening/closing valve **1500**.

(62) In such a case, since the discharge of the filtered water through the outflux **220** from the filtration space **210** is blocked, the production of filtered water using the filter module **100** may be stopped.

(63) Accordingly, the user may temporarily stop the production of filtered water by manipulating the opening/closing valve **1500** if necessary, whereby even if the end of the connection pipe **1300** is separated from the fitting **1220** of the storage tank **1210**, it may be minimized that the filtered water is discarded to the outside from the end of the connection pipe **1300** separated from the fitting **1220**.

(64) In this case, the gravity-type water-purifying device **1000** according to an embodiment of the present invention may further include a filter **1400** to further improve the quality of the filtered water by filtering the filtered water produced by the filtered water production part **1100** again.

(65) Such a filter **1400** may secondarily filter the filtered water produced by the filter module **100**. Accordingly, the filtered water may be secondarily filtered through the filter **1400** and then moved to the filtered water storage space **1230** of the storage tank **1210**.

(66) To this end, the filter **1400** may be provided in the connection pipe **1300** connecting the filtered water production part **1100** and the storage tank **1210** to each other.

(67) For example, the filter **1400** may improve the quality of the filtered water by removing the odor of the filtered water produced through the filter module **100** and/or harmful chemicals included in the filtered water.

(68) As a non-limiting example, the filter **1400** may be a known carbon filter, but the type of the filter **1400** is not limited thereto, and any of the known various filters may be used without limitation as long as it may improve the quality of the filtered water produced from the filter module **100** so that the filtered water may be used as potable water or drinking water. In addition, the filter **1400** may be changed to a suitable type depending on a place where the gravity-type water-purifying device **1000** according to an embodiment of the present invention is installed or used.

(69) Accordingly, the filtered water produced through the filter module **100** in the filtered water production part **1100** may be filtered again while passing through the filter **1400** in the process of moving to the storage tank **1210** side along the connection pipe **1300**.

(70) As a result, the filtered water in a state of being filtered twice through the filter module **100** disposed in the filtration space **210** and the filter **1400** provided in the connection pipe **1300** may be stored in the storage tank **1210**.

(71) Meanwhile, as described above, the filter module **100** may produce the filtered water by filtering out foreign substances included in raw water using gravity.

(72) As such a filter module **100**, any known various filter modules may be used without limitation as long as they may filter out foreign substances from raw water, but preferably may be configured to increase the production of filtered water.

(73) As a non-limiting example, the filter module **100** may include a plurality of plate-shaped filter

members **110** and a filtered water integration member or manifold **120** as shown in FIGS. 5 and 6.
(74) The plurality of plate-shaped filter members **110** may produce the filtered water from the raw water by using water pressure caused by gravity or potential energy in a state in which at least a portion thereof is submerged in the raw water.

(75) In this case, each of the plurality of plate-shaped filter members **110** may be formed in a plate shape having a predetermined area. In addition, the plurality of plate-shaped filter members **110** may be disposed so that one surface of each filter member faces each other at a predetermined interval, and may be spaced apart from each other in parallel along one direction.

(76) Accordingly, since each of the plate-shaped filter members **110** in the filter module **100** may individually produce filtered water, the production of filtered water per unit time may be increased.

(77) In this case, each of the plurality of plate-shaped filter members **110** may be provided in a different form, but may be provided in the same form.

(78) Through this, all of the plurality of plate-shaped filter members **110** in the filter module **100** may be manufactured by using one mold or manufacturing facility, thereby reducing the production cost for producing the plate-shaped filter members.

(79) In addition, the total number of use of the plate-shaped filter member **110** in the filter module **100** may be easily adjusted according to the amount of filtered water required per hour, thereby enhancing the convenience of use.

(80) The plate-shaped filter member **110** may include a support frame **111**, a pair of filtration members **118**, a communication path **116** and a receiving port **117** as shown in FIGS. 7 to 10.

(81) The support frame **111** may support the pair of filtration members **118** and also provide a space for temporarily storing the filtered water produced by the filtration members **118**.

(82) To this end, the support frame **111** may include a loop-shaped edge member **112** having an empty space therein, and a partition member **114** coupled to the edge member **112** so that the empty space may be partitioned into a plurality of storage spaces **115**.

(83) For example, as shown in FIG. 8, the edge member **112** may be a rectangular frame structure with both surfaces open, and the partition member **114** may be a straight member connecting two opposite sides of the edge member **112** to each other.

(84) Further, the partition member **114** may include at least one lateral member **114a** having a predetermined length and at least one longitudinal member **114b** disposed to intersect the lateral member **114a**.

(85) Accordingly, as shown in FIGS. 9 and 10, when the pair of filtration members **118** are attached to both surfaces of the edge member **112**, respectively, the pair of filtration members **118** may be respectively supported by both surfaces of the edge member **112**, and may be maintained in a state spaced apart from each other through the thickness of the edge member **112**.

(86) In this case, the filtration member **118** may be provided in a form of a plate-shaped sheet, and the edge side thereof may be attached to one surface of the edge member **112**.

(87) Accordingly, the filtration member **118** may be maintained in a plate-like shape by the support frame **111**, and may increase the output or flow rate of filtered water by increasing the contact area with the raw water.

(88) However, the shape of the frame member **112** is not limited thereto, and may be appropriately changed to any one of a circular shape, an arc shape, a polygonal shape, and a combination thereof depending on the shape of the filtration member **118**.

(89) In addition, the partition member **114** may be composed of only any one of the lateral member **114a** and the longitudinal member **114b**, and in case any one of the lateral member **114a** and the longitudinal member **114b** is provided in plurality, the plurality of lateral member **114a** or the plurality of longitudinal member **114b** may be arranged to be spaced apart and parallel to each other.

(90) In this case, the partition member **114** may serve to partition the empty space into a plurality of storage spaces **115** and also support the filtration member **118**.

(91) Accordingly, the edge of the filtration member **118** may be supported by the edge member **112**, and the remaining inner region of the filtration member **118** except for the edge thereof may be supported by the partition member **114**.

(92) As a result, when the filtration member **118** is provided as a plate-shaped sheet having a predetermined area, even if water pressure is applied to the filtration member **118** while the raw water passes through the filtration member **118**, the filtration member **118** may be maintained in a plate-like shape without being bent or deformed by the water pressure.

(93) In this case, the edge member **112** may include a stepped portion **113** formed to be recessed inward to a certain depth along the inner side of the edge, and the stepped portion **113** may be formed on each of both surfaces of the edge member **112**.

(94) In this case, as shown in FIG. **8**, a thickness t_2 of the edge member **112** in the portion where the stepped portion **113** is formed may have the same thickness as a thickness t_1 of the partition member **114**.

(95) Accordingly, when the pair of filtration member **118** are respectively attached to both surfaces of the support frame **111**, the edge of the filtration member **118** may be in close contact with the stepped portion **113** as shown in FIG. **9**; the inner region of the filtration member **118** may be supported by the partition member **114**; and the edge portion and inner portion of the filtration member **118** may form a horizontal plane.

(96) The at least one communication path **116** may be formed in the partition member **114**.

(97) As shown in FIG. **11**, the communication path **116** may connect a plurality of storage spaces **115** formed inside the edge member **112** to each other through the partition member **114**.

(98) Accordingly, the filtered water produced while passing through the filtration member **118** may move toward the receiving port **117** via the plurality of storage spaces **115**.

(99) For example, in case the partition member **114** includes both the lateral member **114a** and the longitudinal member **114b**, the communication path **116** may be formed in each of the lateral member **114a** and the longitudinal member **114b**.

(100) In addition, the communication path **116** may be a through-hole formed through the lateral member **114a** and the longitudinal member **114b**, but may also be an accommodation groove formed to be recessed inwardly from one surface of the lateral member **114a** and the longitudinal member **114b**. In this case, the bottom surface of the communication path **116** may face one surface of the filtration member **118**.

(101) That is, when the pair of filtration member **118** are respectively attached to both surfaces of the support frame **111**, the communication path **116** may be formed on each of both surfaces of the partition member **114** so that the bottom surface thereof may face one surface of the pair of filtration members **118**, respectively.

(102) Accordingly, the filtration member **118** may produce the filtered water not only in an area corresponding to the storage space **115**, but also in a portion of the partition member **114** where the communication path **116** is formed.

(103) Accordingly, even if the partition member **114** supports the inner region of the filtration member **118**, an area in which the filtered water cannot be produced may be minimized, thereby further increasing the output of filtered water.

(104) Although the drawings show that the communication paths **116** formed on each of both surfaces of the partition member **114** are formed at positions overlapping each other, the formation position of the communication path **116** is not limited thereto, and the position of the communication path formed on one surface of the partition member **114** may not overlap with the position of the communication path formed on the opposite surface.

(105) The filtration member **118** may be provided in a pair as described above and may be attached to each of both surfaces of the support frame **111**.

(106) The above filtration member **118** may filter out foreign substances contained in the raw water while the raw water to be filtered moves from the outside of the filtration member **118** to the

storage space **115** by gravity or water pressure.

(107) In this case, the filtration member **118** may be provided as a plate-shaped sheet so as to increase the contact area with the raw water.

(108) Herein, although any known filtration member used for water treatment may be applied as the filtration member **118**, the filtration member **118** may also be a plate-shaped sheet comprising a support **118a** and a nanofiber web layer **119b** disposed on one or both surfaces of the support **118a**.

(109) In the present invention, the nanofiber web layer **119b** may filter out foreign substances contained in the raw water, and the support **118a** may support the nanofiber web layer **119b** and serve as a moving passage for diffusing the filtered water produced by the nanofiber web layer **119b** over the entire area of the filtration member **118**.

(110) As a non-limiting example, the filtration member **118** may have a two-layer structure in which the nanofiber web layer **119b** is directly attached to one surface of the support **118a** as shown in FIG. **15(a)**.

(111) Alternatively, the filtration member **118** may have a three-layer structure in which a pair of nanofiber web layers **119b** are directly attached to both surfaces of the support **118a**, respectively, as shown in FIG. **15(b)**. In this case, even if the nanofiber web layer **119b** attached to the outside of the support **118a** among the nanofiber web layers **119b** attached to both sides of the support **118a** is damaged when washing to remove foreign substances attached to the surface of the filtration member **118**, the nanofiber web layer **119b** attached to the inside of the support **118a** may be maintained in its original shape. Accordingly, the nanofiber web layer **119b** may prevent the filtration efficiency of the filtration member **118** from being lowered, and further enhance the quality of the filtered water by filtering the filtered water passing through the support **118a** again.

(112) In this case, the nanofiber web layer **119b** may be laminated to one surface of the support **118a** through thermal welding, ultrasonic welding, high-frequency welding, etc., and one surface of the filtration member **118** may be attached to one surface of the support frame **111** through ultrasonic welding.

(113) In addition, the support **118a** may be a porous substrate so that the filtered water produced through the nanofiber web layer **119b** may be diffused. For example, the support **118a** may be any one of commonly used known woven, knitted, or non-woven fabrics.

(114) Further, the nanofiber web layer **119b** may be formed of nanofibers to filter out foreign substances contained in the liquid to be filtered, and the nanofiber web layer **119b** may include a nanofiber web of a three-dimensional network structure.

(115) In this case, the nanofiber may include a fiber-forming component, including polyacrylonitrile (PAN) and polyvinylidene fluoride (PVDF), and an emulsifier for improving the miscibility of the fiber-forming component.

(116) The nanofiber web layer **119b** may be provided as a single layer or as a multilayer.

(117) The receiving port **117** may discharge the filtered water produced through the filtration member **118** from the plurality of storage spaces **115** to the outside.

(118) To this end, the receiving port **117** may be provided on one side of the support frame **111** to communicate with any one of the plurality of storage spaces **115**.

(119) Accordingly, the filtered water produced through the filtration member **118** may move from each storage space **115** to the storage space **115** in communication with the receiving port **117** through the communication path **116** and then be discharged to the outside through the receiving port **117**.

(120) The filtered water integration member **120** may fix one side of each of the plate-shaped filter members **110** so that the plurality of plate-shaped filter members **110** may be maintained in a state spaced apart in parallel with each other at a distance along one direction while facing each other. Further, the filtered water integration member **120** may integrate the filtered water individually produced from each of the plurality of plate-shaped filter members **110**.

(121) That is, the filtered water integration member **120** may serve as a coupling member for fixing

each of the plurality of plate-shaped filter members **110**, as a spacer member for maintaining a spaced distance between two plate-shaped filter members **110** spaced apart in parallel with each other, and further as a collecting member for integrating the filtered water produced from each of the plate-shaped filter members **110** into one.

(122) To this end, the filtered water integration member **120** may include a body **121**, a plurality of protrusions **123**, a fastening member **125**, a plurality of inlets **126** and at least one outlet **127**, as shown in FIGS. **12** to **14**, and the plurality of inlets **126** may be connected one-to-one with the receiving ports **117** provided in each of the plate-shaped filter members **110** via a tube **130**.

(123) Specifically, the body **121** may include an integration space or interior manifold volume **122** formed therein to integrate the filtered water incoming from each of the plate-shaped filter members **110** through the inlet **126** connected to the tube **130**, and a plurality of inlets **126** connected to the tube **130** may be formed on one side of the body **121**. In addition, at least one outlet **127** for discharging the filtered water integrated in the integration space **122** to the outside may be formed on one side of the body **121**.

(124) Accordingly, the filtered water produced individually from each of the plate-shaped filter members **110** may be moved to the integration space **122** through the receiving port **117**, the tube **130** and the inlet **126**, and the filtered water introduced into the integration space **122** may be discharged to the outside through the outlet **127**.

(125) In this case, the plurality of protrusions **123** may be formed to protrude at a certain height from one surface of the body **121**, and may be disposed to be spaced apart from each other at a predetermined interval.

(126) Accordingly, an accommodation groove **124** into which one side of the plate-shaped filter member **110** is inserted may be formed between two adjacent protrusions **123**.

(127) Herein, the plate-shaped filter member **110** may include a coupler **111a** formed on one side of the support frame **111** to protrude outward by a certain length, and the coupler **111a** may be inserted into the accommodation groove **124**.

(128) Accordingly, when the coupler **111a** provided in each of the plurality of plate-shaped filter members **110** are respectively inserted into the plurality of accommodation grooves **124** formed on one side of the body **121**, the plurality of plate-shaped filter members **110** may be arranged along one direction so that each one surface thereof faces each other, and be maintained in a state spaced apart from each other by the thickness of the protrusions **123**.

(129) As a result, both surfaces of each of the plate-shaped filter members **110** may be in smooth contact with the raw water.

(130) As described above, the filtered water integration member **120** may perform all of a function of integrating the filtered water produced individually through each of the plate-shaped filter members **110** into one, a function of maintaining a spaced distance between two plate-shaped filter members **110** spaced apart in parallel with each other, and a function of fixing each of the plurality of plate-shaped filter members **110**.

(131) Herein, the filtered water integration member **120** may include at least one fastening member **125** that is detachably coupled to the body **121**, and the at least one fastening member **125** may prevent the plate-shaped filter members **110** respectively inserted into the plurality of accommodation grooves **124** from being separated from the accommodation grooves **124**.

(132) For example, the fastening member **125** may be a fastening bar having a predetermined length, and the fastening bar may be fastened to the body **121** to pass through all of the plurality of protrusions **123**.

(133) In this case, a fastening hole **128** may be formed through the plurality of protrusions **123** to allow the fastening member **125** to pass therethrough, and a through-hole **119** may be formed through the coupler **111a** at a position corresponding to the fastening hole **128**.

(134) Accordingly, when the fastening bar is fastened to the body **121** to pass through the fastening hole **128** provided in each of the plate-shaped filter member **110** and the through-hole **119** formed

in each of the plurality of protrusions **123** in a state in which the coupler **111a** provided in each of the plate-shaped filter members **110** is inserted into the accommodation groove **124**, each of the plate-shaped filter members **110** may be fixed to the body **121** in a state in which the coupler **111a** is inserted into the accommodation groove **124**.

(135) Through this, the plate-shaped filter members **110** respectively inserted into the plurality of accommodation grooves **124** may be prevented from being separated from the accommodation groove **124** by the fastening bar.

(136) Accordingly, the total number of use of the plate-shaped filter member **110** fixed to the body **121** in the filter module **100** may be easily adjusted according to the production of filtered water required per hour.

(137) In addition, each of the plate-shaped filter members **110** in the filter module **100** may be detachably coupled to the body **121**, whereby only the plate-shaped filter members requiring replacement among the plurality of plate-shaped filter members **110** may be individually replaced.

(138) Accordingly, the gravity-type water-purifying device **1000** according to an embodiment of the present invention may improve convenience of use and reduce maintenance costs.

(139) That is, since the plurality of plate-shaped filter members **110** and the filtered water integration member **120** in the filter module **100** may be configured as individual modules that may be detached from each other, some destroyed or damaged parts may be individually replaced. Accordingly, the gravity-type water-purifying device **1000** according to an embodiment of the present invention may reduce maintenance costs.

(140) Although an embodiment of the present invention have been described above, the spirit of the present invention is not limited to the embodiment presented in the subject specification; and those skilled in the art who understands the spirit of the present invention will be able to easily suggest other embodiments through addition, changes, elimination, and the like of elements without departing from the scope of the same spirit, and such other embodiments will also fall within the scope of the present invention.

Claims

1. A gravity-flow water purifying device comprising: a housing (**200**) defining an interior volume (**210**), the housing (**200**) having a housing outlet (**220**) at a bottom of the housing (**200**); a filter module (**100**) disposed within the interior volume (**210**), including: a plurality of filter plates (**110**), each plate defining an interior empty space for receiving filtrate; a manifold (**120**) defining an interior manifold volume (**122**), the manifold (**120**) having a plurality of inlets (**126**) in fluid communication with the interior manifold volume (**122**), wherein each inlet (**126**) of the plurality of inlets (**126**) is in fluid communication with the interior empty space of a respective filter plate (**110**), the manifold (**120**) having at least one outlet (**127**) in fluid communication with the housing outlet (**220**), a top of the manifold (**120**) having a plurality of parallel accommodation grooves (**124**) each sized and shaped to receive and support a filter plate (**110**); and a movable storage vessel (**1210**) disposed below the housing (**200**), the movable storage vessel (**1210**) having an inlet opening in fluid communication with the at least one outlet (**127**) for receiving filtered water.
2. The gravity-flow water purifying device of claim 1, wherein the movable storage vessel (**1210**) is box-shaped and includes a fitting (**1220**) formed in the movable storage vessel so as to be detachably coupled to an end of a connection pipe (**1300**).
3. The gravity-flow water purifying device of claim 1, wherein an additional filter (**1400**) is disposed in a connection pipe (**1300**) fluidically disposed between the at least one outlet (**127**) and the inlet of the movable storage vessel for further filtering water exiting the housing (**200**).
4. The gravity-flow water purifying device of claim 3, wherein the connection pipe (**1300**) includes a valve (**1500**) for controlling flow therethrough.
5. The gravity-flow water purifying device of claim 1, wherein each of the plurality of filter plates

(110) includes: a support frame (111) comprising a loop-shaped edge member (112) defining the empty space which is open on both sides, and a partition member (114) coupled to the loop-shaped edge member (112) to partition the empty space into a plurality of storage spaces (115); a pair of filtration members (118) shaped to conform with the shape of the support frame (111), having a predetermined area and being coupled to both surfaces of the support frame (111) to produce filtered water or filtrate from unfiltered water moving from the outside of each filter plate (110) to the plurality of storage spaces (115); at least one communication path (116) formed in the partition member (114) to allow the plurality of storage spaces (115) to be in fluid communication with each other; and a receiving port (117) provided on one side of the support frame (111) and in fluid communication with the plurality of storage spaces (115) and in fluid communication with the interior manifold volume (122) of the manifold (120) via a tube (130) configured to deliver filtered water from the storage spaces (115) to the interior manifold volume (122).

6. The gravity-flow water purifying device of claim 5, wherein the communication path (116) is at least one recessed portion in the partition member (114).

7. The gravity-flow water purifying device of claim 5, wherein the communication path (116) is at least one through-hole in the partition member (114).

8. The gravity-flow water purifying device of claim 5, wherein the edge member (112) includes a stepped portion (113) formed along the inner side of the edge, and an edge side of the filtration member (118) is fixed to the stepped portion (113).

9. The gravity-flow water purifying device of claim 5, wherein the partition member (114) includes at least one lateral member (114a) having a predetermined length and at least one longitudinal member (114b) disposed to intersect the lateral member.

10. The gravity-flow water purifying device of claim 5, wherein the filtration member (118) includes a plate-shaped support (118a) having a predetermined area, and a nanofiber web layer (119b) formed of nanofiber and disposed on one surface or both surfaces of the plate-shaped support (118a).

11. The gravity-flow water purifying device of claim 5, wherein all of the plurality of filter plates (110) comprising filter members made of the same filtration material.

12. The gravity-flow water purifying device of claim 1, further comprising: at least one fastening member (125) removably extending through the structure defining the plurality of parallel accommodation grooves (124) and through a coupler (111a) protruding from each filter plate frame (111), to couple the filter plates (110) to the manifold (120), wherein each coupler (111a) extends into a respective parallel accommodation groove (124).

13. The gravity-flow water purifying device of claim 12, wherein the at least one fastening member (125) is a bar or bolt.
